- Adverse effect in immune system because of bacterial protein.
- Loss of viability of seeds seen in terminator seed technology of GM crops.

### 4.8.7 Polyhydroxybutyrate (PHB)

Synthetic polymers are non-degradable and pollute the soil and when burnt add dioxin in the environment which cause cancer. So, efforts were taken to provide an alternative eco-friendly biopolymers. Polyhydroxyalkanoates (PHAs) and polyhydroxybutyrate (PHB) are group of degradable biopolymers which have several medical applications such as drug delivery, scaffold and heart valves. PHAs are biological macromolecules and thermoplastics which are biodegradable and biocompatible.

Several microorganisms have been utilized to produce different types of PHAs including Gram-positive like *Bacillus megaterium*, *Bacillussubtilis* and *Corynebacterium glutamicum*, Gram-negative bacteria like group of *Pseudomonas* sp. and *Alcaligenes eutrophus*.

### 4.8.8 Polylactic acid (PLA)

Polylactic acid or polylactide (PLA) is a biodegradable and bioactive thermoplastic. It is an aliphatic polyester derived from renewable resources, such as corn starch, cassava root, chips or starch or sugarcane. For the production of PLA, two main monomers are used: lactic acid, and the cyclic diester, lactide. The most common route is the ring-opening polymerization of lactide with metal catalysts like tin octoate in solution. The metal-catalyzed reaction results in equal amount of *d* and polylactic acid.

### 4.8.9 Green Fluorescent Protein (GFP)

The green fluorescent protein (GFP) is a protein containing 238 amino acid residues of 26.9 kDa that exhibits bright green fluorescence when exposed to blue to ultraviolet range (395 nm). GFP refers to the protein first isolated from the jellyfish *Aequorea victoria*. GFP is an excellent tool in biology due to its ability to form internal chromophore without requiring any accessory cofactors, gene products, enzymes or substrates other than molecular oxygen. In cell and molecular biology, the GFP gene is frequently used as a reporter of expression. It has been used in modified forms to make biosensors.



**Figure 4.30:** Polylactic acid product

### 4.8.10 Biopharming

Biopharming also known as molecular pharming is the production and use of transgenic plants genetically engineered to produce pharmaceutical substances for use of human beings. This is also called "**molecular farming or pharming**". These plants are different from medicinal plants which are naturally available. The use of plant systems as bioreactors is gaining more significance in modern biotechnology. Many pharmaceutical substances can be produced using transgenic plants. Example: Golden rice

### 4.8.11 Bioremediation

It is defined as the use of microorganisms or plants to manage environmental pollution. It is an approach used to treat wastes including wastewater, industrial waste and solid waste. Bioremediation process is applied to the removal of oil, petrochemical residues, pesticides or heavy metals from soil or ground water. In many cases, bioremediation is less expensive and more sustainable than other physical and chemical methods of remediation.Bioremediation an eco-friendly approach and can deal with lower concentrations of contaminants more effectively. The strategies for bioremediation in soil and water can be as follows:

- Use of indigenous microbial population as indicator species for bioremediation process.
- Bioremediation with the addition of adapted or designed microbial inoculants.



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